

Toward Industrially Deployable Multi-Agent Reinforcement-Learning Control of Virtual Synchronous Generators

Research Proposal for the Master by Research (MRes)

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1 Introduction

The decarbonisation of electricity supply is replacing conventional synchronous generators with inverter-based renewable resources [1]. Because inverters do not couple a rotating mass to the grid, this shift erodes the physical inertia that damps frequency excursions after a generation–load imbalance. Low-inertia systems then show faster frequency dynamics and larger rates of change of frequency, a recognised threat to stable operation [2, 3, 4].

Virtual synchronous generators (VSGs) are a leading remedy: they program grid-connected power electronics to emulate the electromechanical swing dynamics of a synchronous machine; the emulation exposes a virtual inertia and a virtual damping coefficient that shape the frequency response [5, 6, 7]. In conventional designs these parameters are fixed at commissioning, yet their optimal values depend on the operating point, the load composition, and the disturbance, so a fixed-parameter VSG is sub-optimal under variable renewable-grid conditions [8]. The direct response is to adapt these parameters online and, when several VSGs operate in parallel, to coordinate them across units.

Reinforcement learning (RL) is the dominant data-driven route to such adaptive coordination, and the applicant has built a large body of work mapping what multi-agent RL can and cannot do on this problem (Section 3). In that work the applicant identified a concrete obstacle to real-world use: a learned controller must be trained at length for one specific network, and that training does not transfer to the many topologies a utility operates. The proposed MRes addresses the question this raises: can multi-agent RL control of paralleled VSGs be made topology-general and deployable, effective across different grid configurations without per-network retraining, and where it cannot, when should classical control be preferred?

2 Literature Review

Ernst *et al.* first applied RL to power-system control, comparing fitted-Q iteration against model predictive control [9]; Hadidi *et al.* extended it to wide-area stabilisation [10], and Glavic *et al.* surveyed the field [11]. The continuous-action methods relevant to VSG tuning descend from the deep deterministic policy gradient [12] and its successors, twin-delayed DDPG (TD3) [13] and soft actor–critic (SAC) [14]; coordinating several controllers draws on centralised-training, decentralised-execution multi-agent actor–critic methods [15]. The most direct prior art is Yang *et al.* [16], who govern four paralleled VSGs with independent SAC agents on the modified Kundur four-machine benchmark [17, 18] and report gains over a tuned adaptive baseline (Figure 1). Across this literature one question stays contested: whether the gains of learning-based control justify its opacity and per-network training cost over a well-tuned classical droop controller, which remains the deployed standard [8].

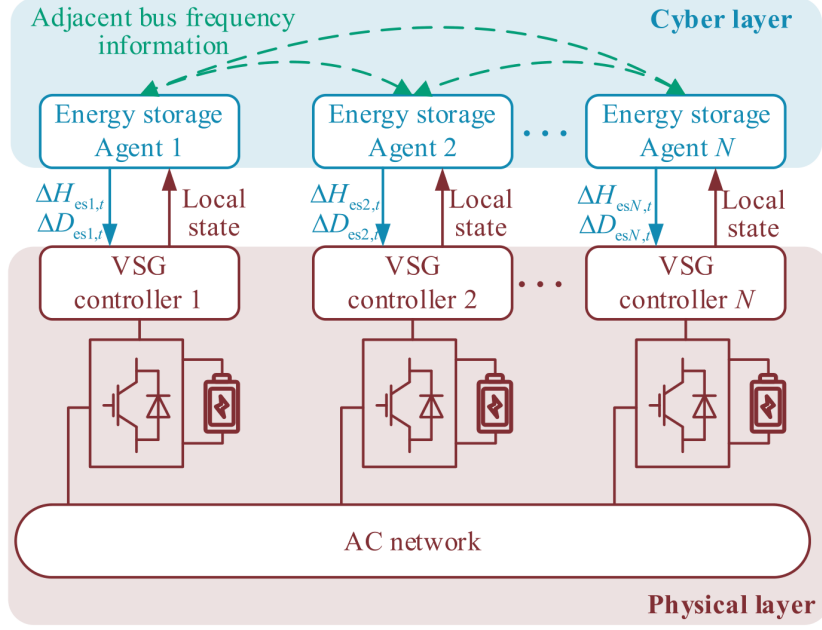


Figure 1: Distributed multi-agent control architecture for paralleled VSGs: each energy-storage agent sets the virtual inertia ΔH and damping ΔD of its converter from local state and neighbouring-bus frequency exchanged over a fault-prone communication layer. Source: Yang *et al.* [16].

The proposed work addresses three gaps. First, deep-RL controllers are sensitive to random seed and early training [19], and reproducibility-aware reporting [20] remains rare in power-system RL. Second, and most important for deployment, these controllers are trained per network: a policy learned on one topology has no principled way to transfer to another, so the training cost recurs for every configuration a utility operates. Third, graph neural networks (GNNs), which operate on the network as a graph, have shown promise for topology-aware power-system learning [21, 22], but have not, to the applicant’s knowledge, been applied to make multi-agent VSG inertia and damping control transferable across topologies. The proposed work sets out to close that gap.

3 Preliminary Work and Demonstrated Capability

The applicant has already carried out extensive prior work on this problem, beginning with a final-year project that reproduced the multi-agent SAC controller of [16] for four paralleled VSGs on the modified Kundur network, using the open-source ANDES phasor-domain simulator [23] as a fully programmable, Python-native training environment. Since then the applicant has extended that work into a sustained research programme¹:

259	experimental rounds (R01–R259)	12	deep-RL algorithm variants
263	audited, traceable findings	91	training trials mapping the plateau
35+	automated regression tests	16-pp	manuscript in IEEE journal format

The experimental framework is a layered, reproducible pipeline:

¹The complete codebase, experiment ledger, and audited claim records are publicly available at <https://github.com/wya5217799/andes-rl-kundur>.

Table 1: The twelve deep-RL controller variants implemented and compared for multi-agent VSG inertia/damping control on the modified Kundur benchmark.

Variant	Function / innovation tested	Outcome
SAC	Reference paper’s multi-agent algorithm [16]	Reproduced; reference behaviour.
SAC-CTDE	Centralised-training, decentralised-execution critic	No gain beyond the plateau band.
TD3	Twin-delayed deterministic policy gradient	Stable deterministic baseline.
TD3-LSTM	Recurrent actor with memory of recent states	Strongest single family; the SOTA base.
TD3-LSTM2	Deeper two-layer recurrence	Within the plateau band.
TD3-LSTM-hreg	Hidden-state regularisation	Improved stability; within band.
TD3-LSTM-warmh0	Warm hidden-state initialisation	Mitigated seed collapse; partial.
TD3-QR-LSTM	Distributional quantile-regression critic	Matched the LSTM SOTA.
TD3-QR-LSTM-hreg	Distributional critic + hidden-state reg.	Within band.
TD3-AFE-LSTM	Adaptive feature-extraction front-end	Within band.
TD3-QR-AFE-LSTM	Combined distributional + adaptive features	Within band.
TD3-Transformer	Attention-based actor over a rolling window	Negative result: deterministic-eval collapse.

- *Scenario layer*: the modified Kundur network and a fixed set of load-disturbance test cases (the LS1/LS2 load steps of [16]), held constant so that every controller faces identical disturbances.
- *Simulation layer*: the ANDES phasor-domain solver [23] wrapped as a reinforcement-learning environment; each training and evaluation step advances a time-domain power-system simulation, so an agent learns against the grid’s own dynamics.
- *Algorithm layer*: the multi-agent controllers of Table 1, interchangeable behind a common interface.
- *Evaluation layer*: two scorers applied to the saved trajectories: the eleven-axis paper-grade metric (the project’s multi-dimensional score) and the paper-faithful cumulative frequency-deviation reward of [16], which places each agent on the reference paper’s own scale.

A single run proceeds train, test, score: an agent is trained online against the simulator; the saved checkpoint is replayed on the fixed disturbance scenarios with learning switched off; and the recorded traces are scored on both metrics and ranked against the no-control and classical-droop baselines.

The applicant implemented twelve distinct controller designs and compared them directly on the same benchmark and evaluation protocol (Table 1), spanning the SAC and TD3 families, recurrent (LSTM) and distributional (quantile-regression) critics, adaptive feature extraction, and a Transformer actor.

(1) The applicant developed a purpose-built eleven-axis paper-grade evaluation metric to expose single-axis failure modes that a scalar reward conceals. (2) Across all twelve variants, single-algorithm performance clustered in a narrow band (roughly 0.38–0.43 on the eleven-axis score; Figure 2); (3) only a heterogeneous-actor weighted ensemble combining several independently trained controllers at inference time broke out of that band, reaching the programme’s best score and roughly a two-fold improvement over a tuned classical adaptive-droop baseline. (4) 91 training trials confirm the band as a structural plateau: a mechanism analysis attributes it to the learned policy acting once and holding

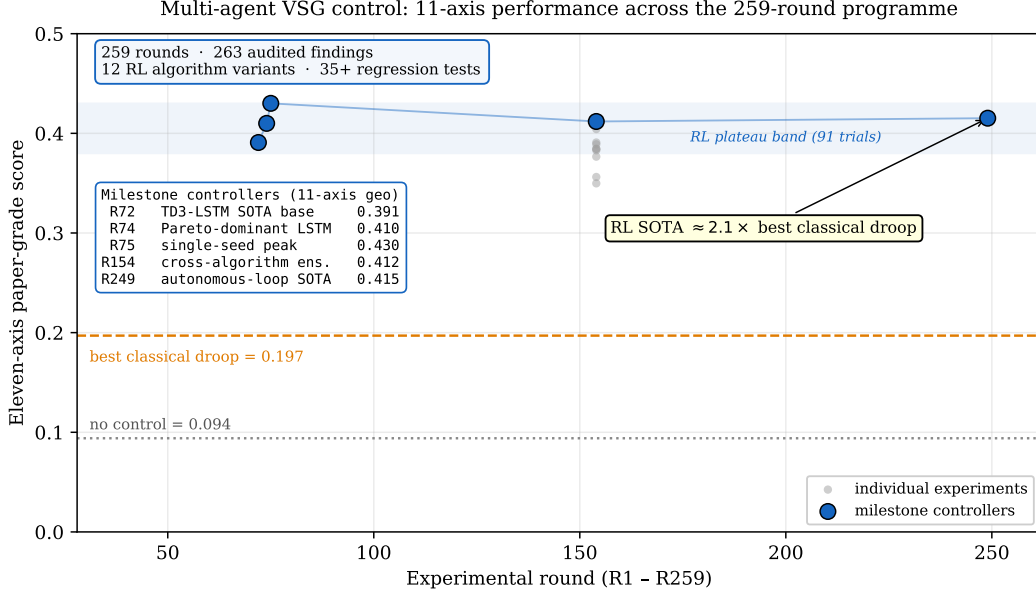


Figure 2: Eleven-axis paper-grade score of the milestone multi-agent controllers across the 259-round research programme, set against the no-control and best classical-droop baselines. The learning-based controllers cluster in a structural performance plateau roughly $2.1 \times$ above the best tuned classical droop controller; every plotted value is pinned to an audited claim in the project ledger (author’s own work).

a near-saturated response decoupled from the ongoing disturbance, whereas classical droop tracks the disturbance proportionally. (5) Reaching a strong controller required this scale of search, 91 trials within 250-plus rounds, on a single fixed topology: a per-network cost that the proposed work addresses.

Figure 3 shows the closed-loop behaviour of the canonical controller: under both load disturbances it holds the four machine frequencies within roughly $\pm 0.05\text{--}0.1$ Hz and returns them toward nominal within a few seconds, while the four agents share the power adjustment.

4 Proposed Research and Methodology

4.1 Aim and research questions

The aim is to move multi-agent RL control of paralleled VSGs from the strong-but-topology-bound position established above toward control that is transferable and deployable. Three research questions arise from those findings:

RQ1 (Necessity). Under what disturbance and operating conditions does learning-based control improve on a well-tuned classical adaptive-droop controller by a margin that justifies its cost, and when is classical control the rational choice?

RQ2 (Limitations). What are the structural limits of per-topology RL agents: the sample cost, the single-network specificity, and the act-and-hold saturation mechanism identified in the preliminary work?

RQ3 (Topology-general deployability, core). Can a graph-neural-network policy, which ingests the power network as a graph, be trained to coordinate VSGs across different grid topologies without per-network retraining? This is the central obstacle to industrial use.

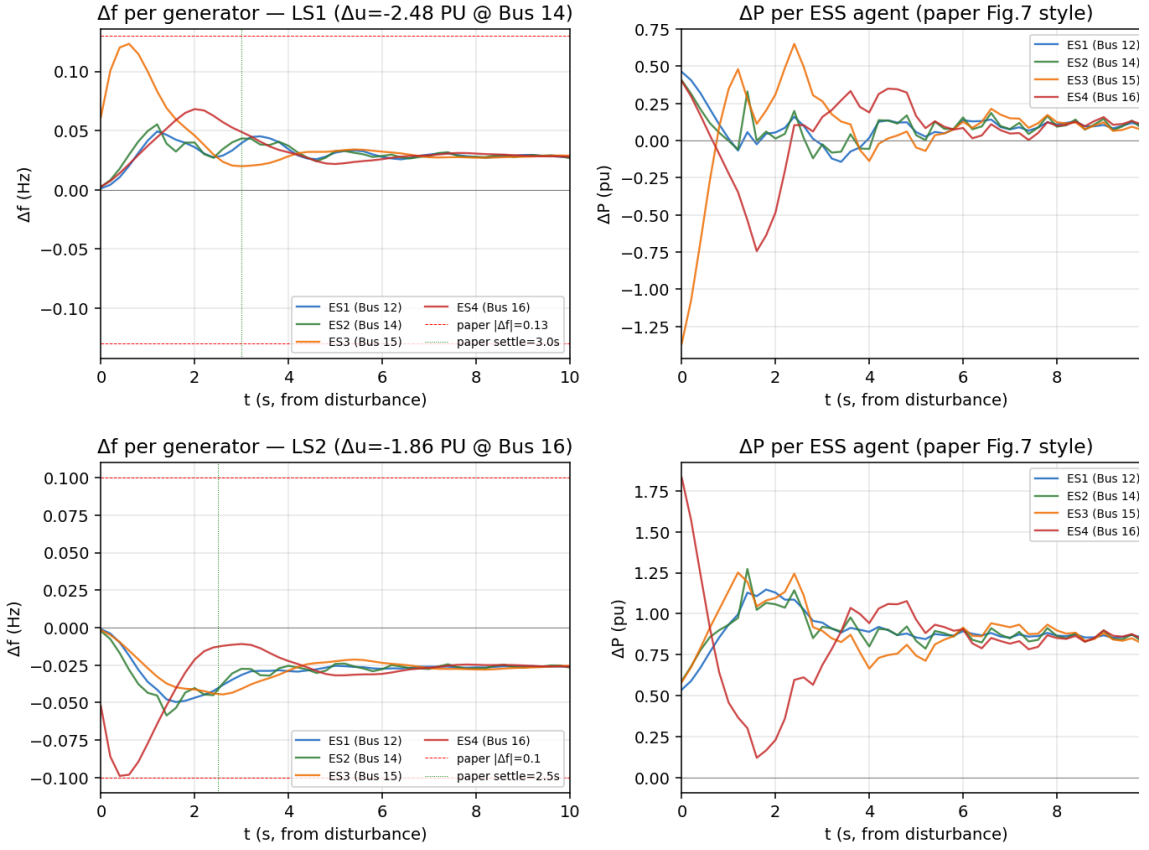


Figure 3: Closed-loop response of the canonical multi-agent controller (R72, eleven-axis score 0.391, four-agent power balance 0.96) under the two load-disturbance scenarios (LS1 top, LS2 bottom). Left: frequency deviation Δf at the four machines; right: active-power adjustment ΔP at the four energy-storage VSG agents. The frequencies return toward nominal within a few seconds and track the reference paper's response (dashed line). Author's own work.

These three questions set the agenda the applicant proposes to pursue; their precise formulation, and the balance among them, will be developed with the supervisor as the work matures, in keeping with the exploratory nature of a research degree.

4.2 Methodology and research approach

The work is computational and draws on three information sources: the ANDES phasor-domain simulator [23], which generates the time-domain frequency responses used for training and evaluation; the VSG-control, RL, and GNN literature; and the applicant's own audited experimental ledger, which supplies verified baselines and guards against regression.

The study will use a set of benchmark topologies: the modified Kundur four-machine system (Figure 4) as the anchor, augmented with additional standard test networks of differing size and structure. Training on a subset and testing on held-out topologies is how transfer will be measured. The advantage is direct relevance to the deployment question; the cost is the effort of building and calibrating several simulated networks, which is scoped as core work for the degree.

The objects under control are the paralleled VSG units and the multi-agent GNN policy that sets their virtual inertia and damping. VSGs are the appropriate subject because they are the locus at which synthetic

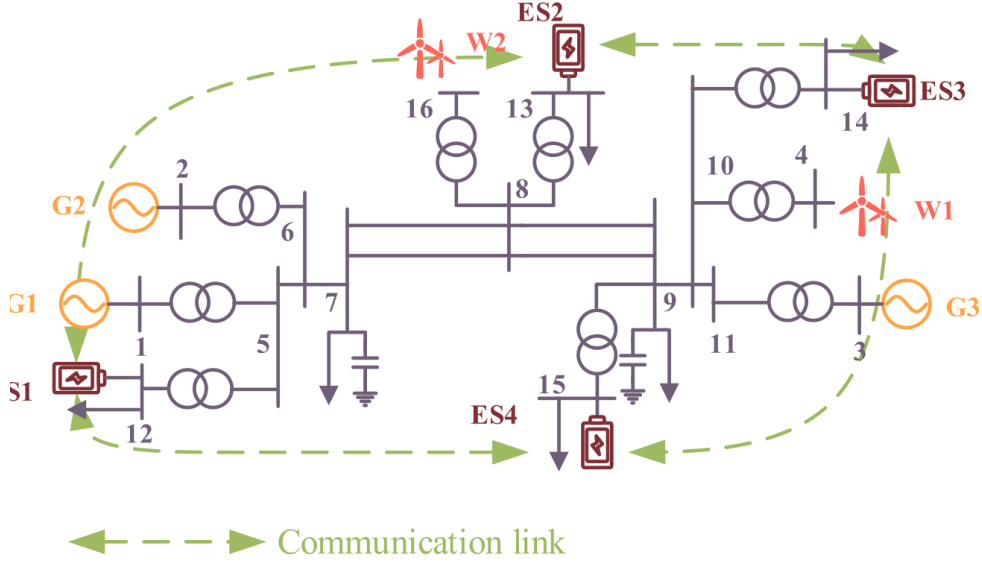


Figure 4: The modified Kundur two-area test system: synchronous generators G1–G3, four energy-storage VSG units (ES1–ES4), wind sources (W1–W2), and the inter-agent communication links. This is the anchor network on which the controllers are trained and evaluated, to be augmented with further topologies to test cross-network transfer. Source: Yang *et al.* [16].

inertia and damping are delivered in an inverter-dominated grid, and their cross-topology coordination is the open problem that fixed-network studies cannot address.

The established ANDES testbed and the eleven-axis, multi-seed evaluation protocol extend to multiple topologies. A GNN-based multi-agent policy will be trained across a family of networks, then evaluated for zero- or few-shot transfer to held-out topologies, and benchmarked against both a tuned classical droop controller (RQ1) and a conventional per-topology RL agent (RQ2/RQ3). Each headline number is reported over a distribution of random seeds, so every claim rests on a reproducible result. The programme is scoped in three phases over the degree: locating the necessity boundary against classical droop (RQ1), quantifying the limits of per-topology agents (RQ2), and building and testing the cross-topology GNN policy (RQ3).

4.3 Rationale for the application

This research continues work already begun under the intended supervisor’s guidance, on an established codebase, evaluation methodology, and experimental record. The MRes provides the supervision, time, and resources to convert a mature exploratory programme into rigorous, publishable research on the deployability question, and is the reason the applicant wishes to undertake this degree at the University of Nottingham Ningbo China.

5 Anticipated Contribution

The research offers three contributions. The first is a topology-general, GNN-based multi-agent controller for VSG inertia and damping support that transfers across grid configurations without per-network retraining, the principal obstacle to deploying learning-based VSG control. The second is a cost–benefit account of when learning-based control justifies its complexity over well-tuned classical methods, which

existing studies rarely provide. The third is a reproducible, multi-dimensional evaluation framework, developed in response to the documented reproducibility shortcomings of deep RL [19, 20] and extended here to the cross-topology setting. All three address a practical deployment question for future low-inertia grids, where grid codes require fast frequency response [24] and the safety of autonomous adaptive control remains open [25].

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